KuaiLive-M3 live-viewing data card

Descriptive statistics and a field dictionary for the raw files this project uses. Source: Hugging Face `imgkkk2004/KuaiLive-M3`, revision `417923d442a7`. All numbers are computed over the full released files (not only the project cohorts) by `docs/data_card/compute_stats.py`. No row-level data are shown.

> The upstream repository has no data card or field-level documentation. The KuaiLive-M3 paper (arXiv:2607.24862) lists the column names and describes user sampling and how the segment embeddings were built. All other meanings below are read from the field names (*by name*) or derived from the data (*inferred*), with the supporting statistic next to them.

21,938
viewers (sampled active users)

35.8M
visit records

6.56M
broadcasts (rooms)

1.06M
streamers

28 days
4–31 March 2026 (UTC+8)

45%
of records last < 5 s

1. Files

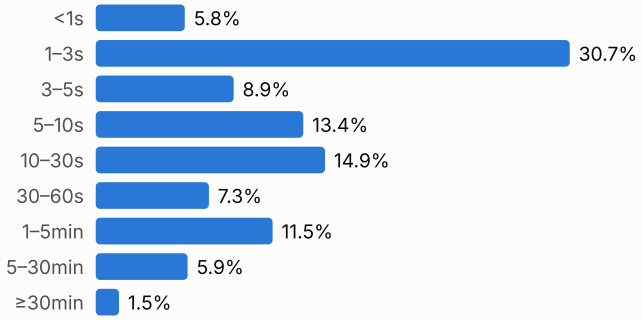
File	Rows	One row is	Used here
`live/live_interaction.csv`	35,806,596	One viewing record: a user in one broadcast, with per-record action counts	Visits, labels, history, entry state
`live/live_room_meta.parquet`	6,564,013	One broadcast with start/end time	Broadcast age, broadcast-end rule
`live/live_like.csv`	49,540,590	One timestamped like event	Like counts in past windows
`live/live_comment.csv`	8,281,923	One timestamped comment	Comment counts in past windows
`live/live_emb_128_ts/` (18 parts)	88,223,794	One content segment of a broadcast with a 128-d embedding	Latest past content embedding
`user_id_set.csv`	21,938	One released user ID	User universe

Not used: live impressions, shares, questionnaire, room-level embeddings, author profiles and the short-video domain. Note that event files are not one-to-one with the per-record counters: `live_like.csv` has 49.5M events while `like_cnt` sums to 198.7M; `live_comment.csv` has 8.3M rows while `comment_cnt` sums to 20.0M. Treat them as separate measurements.

2. Viewing records (`live_interaction.csv`)

Watch duration

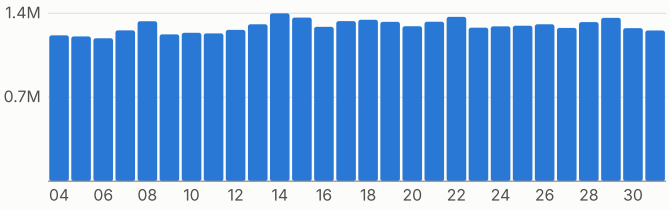
Very short records dominate: median 6.2 s, 45.4% under 5 s, 73.7% under 30 s; only 4.5% reach 10 min. This is why the project uses a fine time grid in the first seconds.



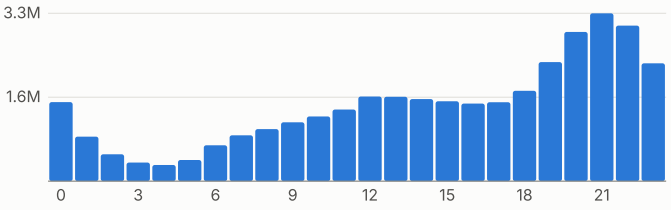
Share of all 35,806,596 records in each duration range (unequal widths).

Quantile	play_duration
P10	1.2 s
P25	1.9 s
P50	6.2 s
P75	33.7 s
P90	187.1 s
P95	524.2 s
P99	2,456.7 s
P999	6,807.5 s
MAX	10,000.0 s

Time coverage



Visit records per partition day (p_date, March 2026, UTC+8). Hover a bar for the exact count.



Visit records by local start hour (UTC+8).

Per viewer

The median viewer has 753 records (P10 111, P90 3,915; max 67,865), is active on 28 of 28 days, watches 189 distinct streamers and 30 h in total (P90 138 h). Heavy users contribute many records, so record-level statistics are dominated by them.

Field dictionary

Field	Type	Meaning	Statistics	Use in this project
`user_id`	int64	Anonymized viewer ID; the released universe is listed in `user_id_set.csv`.	21,938 distinct; 0 NULL	Grouping; cohort assignment
`author_id`	int64	Streamer (creator) ID.	1,061,343 distinct	Per-streamer history
`live_id`	int64	One broadcast (room instance). Joins `live_room_meta.live_id`; a streamer has many.	6,564,013 distinct; 100.0% matched to room meta (author consistent 100.0%)	Room key
`live_play_start_timestamp`	timestamp (ms)	Watch start. Stored without zone; values are UTC (inferred: p_date equals the UTC+8 calendar date).	2026-03-03 16:00:00 → 2026-03-31 15:59:59.999 (UTC) = 4–31 March 2026 local	Visit reconstruction
`live_play_end_timestamp`	timestamp (ms)	Watch end. Records crossing local midnight are cut at `23:59:59.999` and continue in a new record.	end − start equals play_duration within 5 s for 100% of records	Labels (future information; never a feature)
`p_date`	string YYYYMMDD	Partition date in UTC+8.	28 days, 20260304–20260331	—
`live_source_category`	string	Page the user came from (see values below).	21 values; 0 NULL	Feature (one-hot, entry state)
`enter_live_action`	string	How the user entered (see values below).	8 values; 0 NULL	Feature (one-hot, entry state)
`is_auto_play`	0/1	Stream started playing automatically.	8.0% of records are 1	Feature
`is_follow_enter`	0/1	User already followed the streamer when entering (by name); not the same as	61.1% of records are 1	Feature

Field	Type	Meaning	Statistics	Use in this project
		coming from the follow page.		
`is_follow_leave`	0/1	Follow status at leave time.	61.6% of records are 1	Not used (known only at exit)
`play_duration`	int64 ms	Watch duration of the record.	median 6.2 s; P90 187 s; P99 41 min; max 10,000 s (values stop just below 10^7 ms, possibly a cap)	Labels; history features
`like_cnt`	int64	Likes sent during the record.	>0 in 3.54% of records; mean 5.549; max 94,685	Feature (past visits only)
`comment_cnt`	int64	Comments posted during the record.	>0 in 4.14% of records; mean 0.557; max 100,472	Feature (past visits only)
`send_gift_cnt`	int64	Gift-sending actions during the record.	>0 in 0.97% of records; mean 0.042; max 2,408	Feature (past visits only)
`send_gift_num`	int64	Number of gift items sent.	>0 in 0.96% of records; mean 0.238; max 62,702	Not used
`follow_author_cnt`	int64	Follow actions on the streamer.	>0 in 0.66% of records; mean 0.007; max 257	Not used
`cancel_follow_author_cnt`	int64	Unfollow actions.	>0 in 0.25% of records; mean 0.003; max 48	Not used
`share_cnt`	int64	Shares of the stream.	>0 in 0.27% of records;	Not used

Field	Type	Meaning	Statistics	Use in this project
			mean 0.006; max 241	
`report_live_cnt`	int64	Reports of the stream.	>0 in 0.06% of records; mean 0.002; max 294	Not used

No field has NULL values. 755 records are exact duplicates of another record. Action counters describe the whole record, so they are used only for *completed* past visits, never for the current visit.

Entry values

`enter_live_action` : most entries are swipes in the vertical feed or taps.

Value	Records	Share		Meaning / evidence
`SLIDE_UP_ENTER`	18,963,902	52.96%		Swiped up into the room.
`CLICK_ENTER`	13,023,223	36.37%		Tapped into the room.
`SLIDE_DOWN_ENTER`	2,455,227	6.86%		Swiped down into the room.
`OTHER`	591,721	1.65%		Other action; not documented.
`UNKNOWN`	465,230	1.30%		Unknown (a string value, not NULL).
`SCHEMA_ENTER`	169,418	0.47%		Probably a deep-link/route jump (inferred).
`AUTO_ENTER`	108,846	0.30%		Automatic entry; not identical to is_auto_play.
`RETURE_REFER_LIVE`	29,029	0.08%		Probably a return to a referred stream (inferred; source spelling kept).

`live_source_category` : more than half of all records come from the following page.

Value	Records	Share		Meaning / evidence
`follow`	19,951,707	55.72%		Following page (by name). Different from is_follow_enter.
`featured_feed`	5,979,447	16.70%		Featured feed (by name).
`find_feed`	3,493,539	9.76%		Discover feed (by name).
`nearby`	2,069,383	5.78%		Nearby page (by name).
`other`	1,863,755	5.21%		Other sources; grouping rule not published.
`search`	587,008	1.64%		Search (by name).
`find_tab`	553,140	1.54%		Discover tab (by name).
`detail_head`	468,699	1.31%		Unclear; possibly a detail-page header.
`live_more_square`	296,587	0.83%		‘More live’ square (by name).
`fanstop_award`	157,477	0.44%		Business code; meaning not documented.
`profile`	157,402	0.44%		Profile page (by name).
`live_end`	116,239	0.32%		Stream-ended page (by name).
`live_square`	39,120	0.11%		Live square (by name).
`app_out_push`	25,517	0.07%		Push notification related (by name).
`live_jump`	17,260	0.05%		Live-to-live jump (by name).
`live_out_push`	13,487	0.04%		Push related (by name).
`home_live_square`	11,712	0.03%		Home live square (by name).
`live_push`	3,160	0.01%		Push related (by name).
`featured_live_square`	1,664	0.00%		Featured live square (by name).
`kuaishou_coin_task`	228	0.00%		Coin-task entry (by name).
`arrow`	65	0.00%		Not documented.

The model keeps categories with at least 1,000 visits in cohort A's training period and maps the rest to “other” (vocabulary: `configs/entry_category_vocab.json`).

3. Broadcasts (`live_room_meta.parquet`)

Field	Type	Meaning	Statistics	Use in this project
`live_id`	int64	Broadcast ID (unique key).	6,564,013 rows = distinct	Join key
`author_id`	int64	Streamer ID.	1,061,343 distinct	Consistency check
`live_name`	string	Broadcast title (free text).	NULL in 11.4%	Not used, not shown
`start_timestamp`	timestamp (ms, UTC)	Broadcast start. Starts before the window are kept (earliest 2026-02-24).	0 NULL	Broadcast age feature
`end_timestamp`	timestamp (ms, UTC)	Broadcast end T_b. NULL when the stream was still live at data end.	NULL in 26,888 rows (0.41%); never before start	Exit-cause rule (broadcast vs user)
`live_duration`	double ms	end − start.	median 2.1 h; P90 5.9 h	—

Every viewing record joins exactly one broadcast with the same streamer. 1.2% of records end more than 5 s after their broadcast's recorded end; the sample protocol handles these explicitly.

4. Like and comment events

`live_like.csv`

Field	Type	Meaning	Statistics	Use
`user_id`	int64	Liker.	20,483 distinct	Event features
`live_id`	int64	Broadcast.	568,262 distinct	Room join
`like_timestamp`	timestamp (ms, UTC)	Time of the like event.	2026-03-03 15:59 → 2026-03-31 15:59	Windows [t−w, t)

Median liker has 164 like events (P90 3,729).

`live_comment.csv`

Field	Type	Meaning	Statistics	Use
`live_stream_id`	int64	Broadcast (same key as `live_id`).	668,995 distinct	Room join
`user_id`	int64	Commenter.	20,521 distinct	Event features
`author_id`	int64	Streamer.	—	—
`content`	string	Comment text (free text, Chinese). 4.4% are the placeholder `[表情评论]` (emoji comment).	median length 6 chars	Not used, not shown
`comment_ts`	int64 epoch ms	Time of the comment.	2026-03-03 15:59 → 2026-03-31 15:59 (UTC)	Windows [t−w, t)

5. Content segments (`live_emb_128_ts/`)

Field	Type	Meaning	Statistics	Use
`live_stream_id`	string	Broadcast ID (as text).	≈6,108,894 distinct	Room join
`segment_id`	string	Segment identifier.	—	Audit
`feature_128`	list<float32> [128]	Segment embedding from an industrial multimodal LLM over clips, ASR text and metadata, reduced to 128 dims with PCA (paper). Dimensions have no individual meaning.	always 128 dims	Content features
`timestamp`	int64 epoch ms	Segment time; this project treats it as the moment the segment becomes available.	2026-03-03 15:59 → 2026-03-31 15:59; spacing within a stream median 66 s (P10 32 s, P90 36 min)*	Past-only alignment

* Spacing and segments per stream (median 5, P90 27) are from a 1% hash sample of streams across all 18 parts; parts are not split by stream. The upstream model and PCA training window are not documented, so the embeddings could contain information from later in the month.

6. Known caveats

- **Day partitioning.** Records crossing local midnight are cut into two records (end `23:59:59.999`, restart `00:00:00.000`). Without stitching they create false exits and entries.
- **Overlapping records.** One user can have overlapping records in different rooms (e.g. background playback). Which room the user was watching is ambiguous during the overlap.

- **Broadcast end.** Records of viewers who stayed to the end close a few seconds before or after the broadcast end, so “record ended” is not always a user decision.
- **Sampled audience.** Only ~22k sampled users are released; other viewers in a room are not observed.
- **Duration cap.** No record exceeds 10^7 ms (about 2.8 h); longer views may be split or truncated.

How the project turns these records into visits and exit labels: `docs/sample_protocol.md`.

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7. Cohorts used by this project

All released users are ordered by SHA-256 of their ID. Cohort A (1,000 users) is used for training (first 18 days) and validation (next 5 days). Cohorts B (300) and C (1,000) are disjoint from A and drawn from the 21,748 users with at least one record in the last five days, which is their test period. The project therefore uses 2,300 of the 21,938 users; cohort hashes are recorded in `results/formal-v1.1.0/feature_manifest.json`.

License: the dataset page lists CC BY-SA 4.0 while the paper states CC BY-NC-SA 4.0; treat as non-commercial research use; cite arXiv:2607.24862. Generated from `stats.json` (computed in 69 s with DuckDB).